Stormwater Regulation Summary: Ohio (Statewide)

Municipal Stormwater Management Technical Profile | Sempergreen USA

1. Regulatory Overview

- Primary driver: NPDES MS4 permit compliance and Ohio EPA post-construction stormwater management.
- Ohio EPA Construction General Permit (CGP) governs projects with ≥1 acre of soil disturbance.
- Local MS4 permittees (cities, counties) often have stricter requirements than state minimums.
- Post-construction BMP required for all qualifying developments — permanent stormwater management.
- Ohio EPA Water Resource Restoration Sponsor Program (WRRSP) provides incentives for green infrastructure.

2. Typical Soil Conditions

- Ohio soils vary significantly by region — glaciated (north/central) vs. unglaciated (southeast).
- Glaciated areas: heavy clay tills (Hoytville, Pewamo series) in NW; silty clay loam in central (Crosby, Celina).
- Unglaciated SE Ohio: Gilpin-Upshur-Wellston complex — silt loam over shale, moderate to poor drainage.
- Hydrologic Soil Group: Predominantly C-D across glaciated areas; mixed B-C in unglaciated regions.
- Implication: Statewide, many soils have limited infiltration — detention and retention approaches often more practical than infiltration.
- *Soil conditions are highly variable by county — site-specific investigation required.*

3. Green Roof Requirements

- No statewide green roof mandate — Ohio does not have a green roof ordinance at any level of government.
- Ohio EPA accepts green roofs as an approved post-construction BMP under the CGP.
- Local stormwater fee credit programs (Columbus, Cleveland, Cincinnati) incentivize green roofs.
- Ohio PACE (Property Assessed Clean Energy) financing can be used for green roof and solar installation.
- Market is developing — green roof awareness is lower than coastal cities but growing, especially in Columbus and Cleveland.

4. TSS Requirements

- Ohio EPA requires 80% TSS removal for post-construction stormwater management.
- Green roofs provide TSS removal credit for captured rainfall volume — accepted in Ohio EPA BMP manual.
- WQv (Water Quality Volume) capture of the 0.90" event inherently addresses the highest-TSS first flush.
- Additional TSS treatment may be required for ground-level impervious not managed by rooftop BMPs.
- *TSS credit rates for green roofs vary by local MS4 authority — check specific municipality.*

5. Nature-Based Retention Requirements

- WQv: Retain 0.90" per Ohio EPA CGP statewide standard; some local MS4 communities may require more.
- Ohio EPA accepts volume-based retention or treatment-based approaches for WQv compliance.
- CN (Curve Number) method per NRCS TR-55 is the standard statewide design approach.
- Critical Storm Method: Some Ohio communities require sizing detention to control the "critical storm" — the storm whose post-development peak most exceeds pre-development.
- Typical urban CN: 85-98; with green roof BMP: 65-75.
- *Requirements vary significantly by MS4 authority — confirm with specific municipality.*

6. Allowable Outflow Rates

- Critical Storm Method: Control post-development peak to ≤ pre-development for the critical storm event.
- 1-year extended detention: 24-hour drawdown of the 1-year, 24-hour storm (common in many Ohio communities).
- Ohio does not have a statewide flat l/s/ha rate — requirements are relative to pre-development conditions.
- For reference, typical undeveloped Ohio site (CN 65-70): approximately 15-25 l/s/ha for the 10-year storm.
- *Check specific MS4 community requirements — Columbus, Cleveland, and Cincinnati each have distinct standards.*